2. Find the tolerance value of the resistor given below

Green Blue Red Silver
5 6 X 10^2 ± 10%
5600 Ω ± 10% i.e., 560 Ω

Tolerance value of 5.6 kΩ = 5040 Ω to 6160 Ω

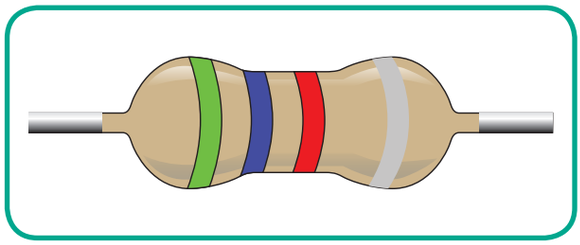


Figure 1.45(b) Calculating tolerance code

1.9.2 Faults in Resistors

1. Resistors may get opened because of overheat due to excess current flow.
2. Resistance value becomes higher than the original value.
3. Faulty resistors cannot be used in circuits.

1.9.3 Testing of resistors

1. Using multimeter, test a resistor, If the dial remains unmoved, then the resistor is open
2. If meter shows higher value than that of the actual value of the resistor, the resistor becomes high value.

1.10 KIRCHHOFF'S LAWS

It is used to find out the current flow in the network circuits easily where it is not easy to find values using ohm's law. It is applicable for both DC and AC circuits. There are two types of law viz.

1. Current law
2. Voltage law



German scientist Robert Kirchchoff created these laws and hence called as Kirchhoff's laws.

1.10.1 Current law

The sum of the currents flowing towards a junction is equal to the sum of the currents flowing away from it. This is called Kirchhoff's current law.

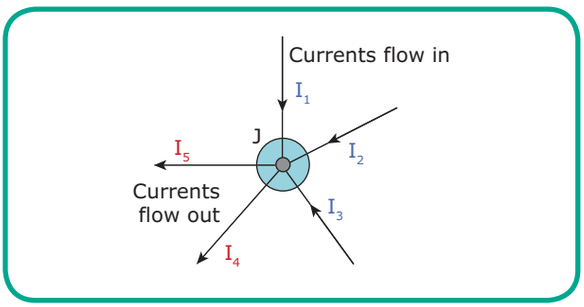


Figure 1.46 Kirchhoff's current law

In the Figure 1.46, J is the junction (or node) formed by five conductors. The current in these conductors are I_1, I_2, I_3, I_4 and I_5. Here, I_1, I_2 *and* I_3 flow towards the junction and I_4 *and* I_5 flow away from the junction. According to Kirchhoff's current law,

$$I_1 + I_2 + I_3 = I_4 + I_5$$

i.e. currents flow towards junction = currents flow away from junction.

On the other hand, $I_1 + I_2 + I_3 - I_4 - I_5 = 0$. This is known as Kirchhoff's current law equation.

1.10.2 Voltage law

At any closed circuit, the sum of potential drops across each resistor in a series circuit is equal to the supply voltage given to the circuit. The circuit representing the voltage law is shown in Figure 1.47.

ELEC_EQUIP_Chapter01.indd 23 11/10/2020 12:22:44 PM